

EXPERIMENT: 29
CREATING THE APPLICATIONS USING TCP ECHO SERVER AND CLIENT IN
JAVA/C

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <sys/wait.h>

int main()
{
    int server, client, sock;
    struct sockaddr_in addr;
    char msg[100];
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        /* SERVER */

        server = socket(AF_INET, SOCK_STREAM, 0);

        addr.sin_family = AF_INET;
        addr.sin_port = htons(8080);
        addr.sin_addr.s_addr = INADDR_ANY;
```

```

bind(server, (struct sockaddr *)&addr, sizeof(addr));

listen(server, 1);

printf("Echo Server waiting...\n");

client = accept(server, NULL, NULL);

recv(client, msg, sizeof(msg), 0);

printf("Client Message: %s\n", msg);

send(client, msg, strlen(msg) + 1, 0);

close(client);
close(server);
}
else
{
    /* CLIENT */

    sleep(1);

    sock = socket(AF_INET, SOCK_STREAM, 0);

    addr.sin_family = AF_INET;
    addr.sin_port = htons(8080);
    addr.sin_addr.s_addr = inet_addr("127.0.0.1");

```

```
connect(sock, (struct sockaddr *)&addr, sizeof(addr));

printf("Enter message: ");
fgets(msg, sizeof(msg), stdin);

msg[strcspn(msg, "\n")] = '\0';

send(sock, msg, strlen(msg) + 1, 0);

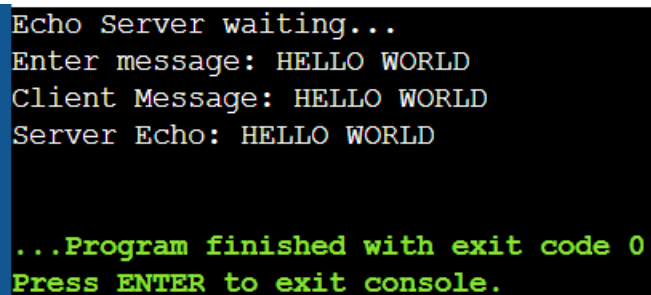
recv(sock, msg, sizeof(msg), 0);

printf("Server Echo: %s\n", msg);

close(sock);

wait(NULL);
}

return 0;
}
```

A terminal window with a black background and a blue vertical bar on the left. The text is displayed in a monospaced font. The first four lines are in white: 'Echo Server waiting...', 'Enter message: HELLO WORLD', 'Client Message: HELLO WORLD', and 'Server Echo: HELLO WORLD'. The next two lines are in green: '...Program finished with exit code 0' and 'Press ENTER to exit console.'

```
Echo Server waiting...
Enter message: HELLO WORLD
Client Message: HELLO WORLD
Server Echo: HELLO WORLD

...Program finished with exit code 0
Press ENTER to exit console.
```